

LAB EXPERIMENT

Name: Dhinakaran. S

Register Number: 192412352

Course Code: ECA0901

Course Name: DIGITAL SIGNAL PROCESSING

3. Performance Analysis of IIR Filters

3.1 Design and Comparative Analysis of Butterworth and Chebyshev Type-I IIR Band-Pass Filters Using MATLAB

CODE:

```
clc;
clear;
close all;
% Specifications
Fs=2000; Rp=1; Rs=40;
Wp=[300 600]/(Fs/2);
Ws=[200 700]/(Fs/2);

% Butterworth Filter
[n1,W1]=buttord(Wp,Ws,Rp,Rs);
[b1,a1]=butter(n1,W1,'bandpass');

% Chebyshev Type-I Filter
[n2,W2]=cheb1ord(Wp,Ws,Rp,Rs);
[b2,a2]=cheby1(n2,Rp,W2,'bandpass');

% Magnitude Response
figure;
freqz(b1,a1); title('Butterworth Filter');

figure;
freqz(b2,a2); title('Chebyshev Type-I Filter');

% Group Delay
figure;
```

```

grpdelay(b1,a1); hold on;
grpdelay(b2,a2);
legend('Butterworth','Chebyshev-I');

% Pole-Zero Plot
figure;
subplot(121),zplane(b1,a1),title('Butterworth')
subplot(122),zplane(b2,a2),title('Chebyshev-I')

% Impulse Response
figure;
subplot(211),impz(b1,a1),title('Butterworth Impulse')
subplot(212),impz(b2,a2),title('Chebyshev Impulse')

% Pole Magnitude
figure;
subplot(121),stem(abs(roots(a1)),'filled')
title('Butterworth Pole Magnitude')

subplot(122),stem(abs(roots(a2)),'filled')
title('Chebyshev Pole Magnitude')

disp(['Butterworth Order = ',num2str(n1)])
disp(['Chebyshev Order = ',num2str(n2)])

```

OUTPUTS:

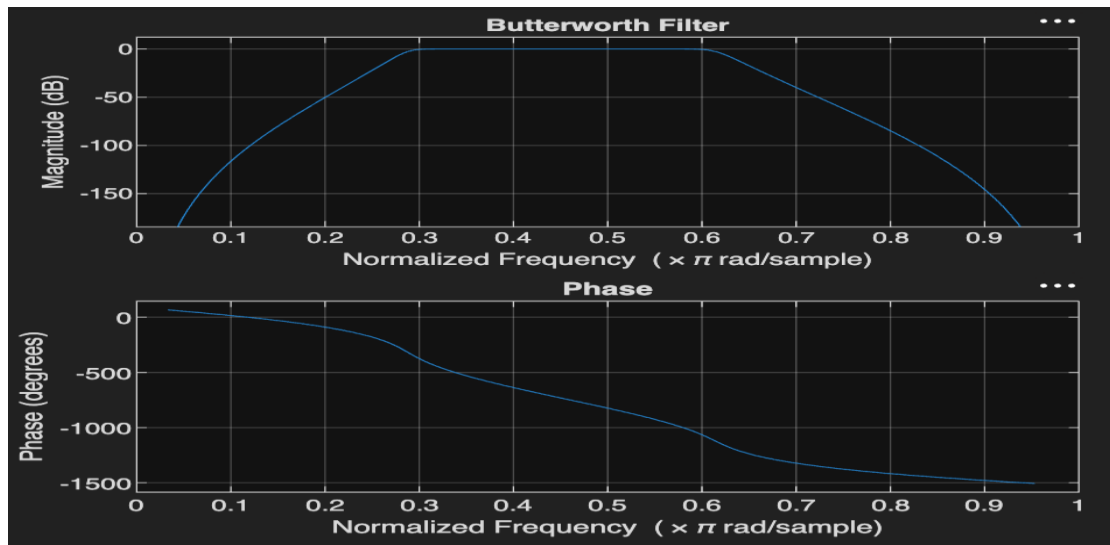
Objective 1:

```

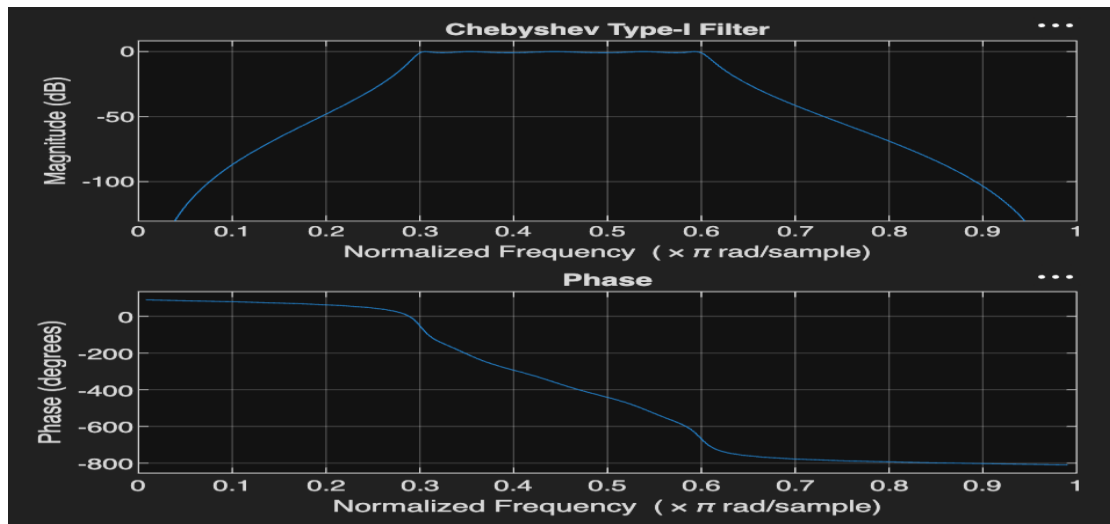
Butterworth Order = 9
Chebyshev Order = 5

```

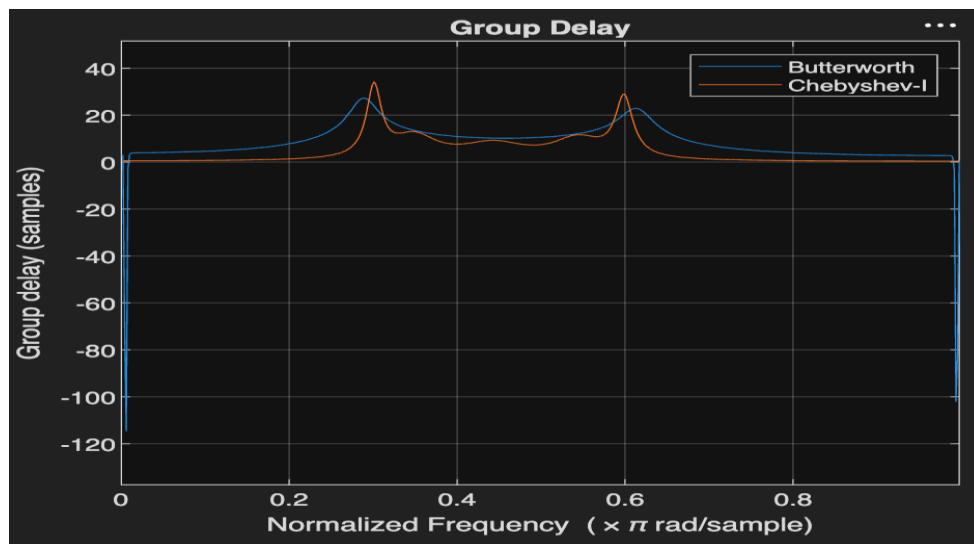
Objective 2:



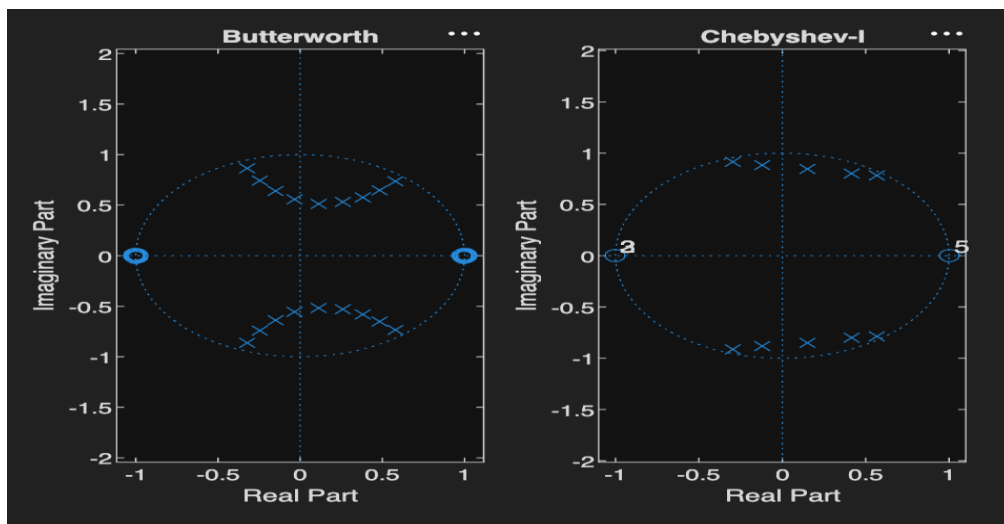
Objective 3:



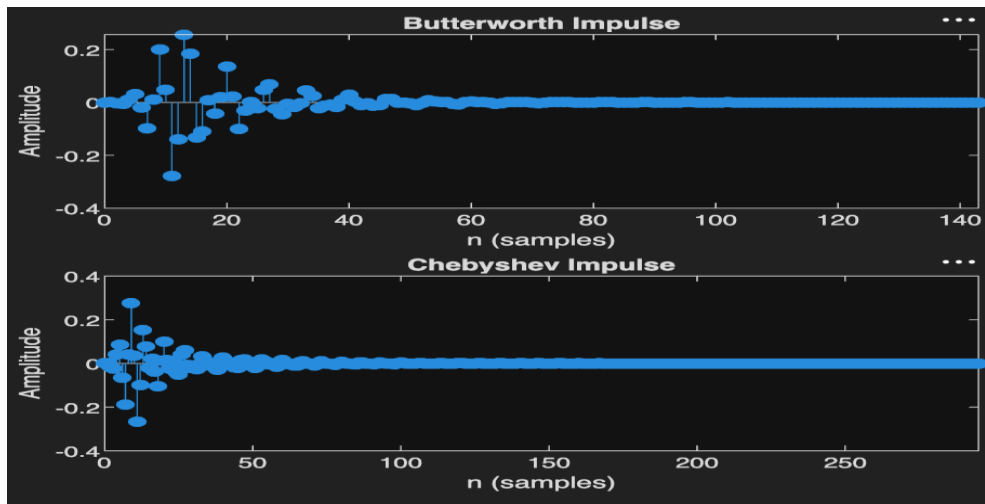
Objective 4:



Objective 5:



Objective 6:



Objective 7:

